

Yoann SABATIER MONTANARO

sabatieryoann@gmail.com · yoannsab.github.io/portfolio · linkedin.com/in/yoann-sabatier-montanaro · github.com/YoannSab

AI/ML Software Engineer graduated from INSA Lyon and Polytechnique Montréal (Mila ecosystem). Background combining Deep Learning research, operational optimization, and mobile/fullstack application development. Proven track record in processing complex datasets, designing secure software.

TECHNICAL SKILLS

ML & AI: PyTorch, Scikit-Learn, Pandas, Hugging Face, vLLM, LangChain, Representation Learning, NLP, Time-Series
Software & Mobile: Python, C++, TypeScript, React Native, ReactJS, Node.js, Linux OS, Git, CI/CD, Docker, Security
Languages: French (Native), English (C1 – Professional Proficiency), Italian (B1)

EDUCATION

Master of Science (M.Sc.A.) in Computer Science – AI & Operations Research 2023 – 2025
Polytechnique Montréal (Mila Ecosystem) Montréal, QC

- **GPA: 4.0 / 4.0.** Core focus: Representation Learning (Pr. Aaron Courville, Mila), Integer Linear Programming, Metaheuristics, and Constraint Programming. /
- Graduated with Highest Honors

Master's Degree in Computer Science and Software Engineering 2020 – 2025
INSA Lyon Lyon, France

- Academic Excellence: Ranked **4th out of 130** students in the Computer Science Department.
- Graduated with Highest Honors

PUBLICATIONS

Sabatier Montanaro, Y., Jacquet, T. J., Cappart, Q., & Desaulniers, G. (2025). *"A Column Generation Heuristic for Multi-depot Electric Bus Scheduling"*. Proceedings of the International Conference **CPAIOR 2025** (Melbourne), published in Springer LNCS.

- Developed a hybrid framework (Metaheuristic & Operational Research) to solve large-scale electric transit network scheduling under physical, temporal, and energy constraints.

PROFESSIONAL EXPERIENCE

Mobile & Fullstack Software Engineer Sep 2025 – Present
onepoint Paris, France

- Built and deployed a secure mobile banking application using React Native from scratch to launch in 8 months.
- Ensured full compliance with banking security standards and optimized app performance for end-users.

Graduate Research Assistant / Machine Learning Jan 2024 – Apr 2025
Corail Research Group – Polytechnique Montréal & Mila Montréal, QC

- Designed a novel graph-based framework combining Metaheuristic and Operations Research for transit fleet optimization, achieving a **35% reduction in computation time**.
- Served as **Teaching Assistant** for *INF8175: Introduction to AI* (Pr. Quentin Cappart), mentoring Bachelor, Master, and Ph.D. students during lab sessions and projects.

Data Scientist Intern May 2023 – Aug 2023
onepoint Paris, France

- Designed and evaluated Deep Neural Network architectures dedicated to complex time-series classification.

Research Intern Jun 2022 – Aug 2022
Inria Lyon (Beagle Team) Lyon, France

- Built and documented a Python library for 3D volumetric brain image segmentation (Allen Brain Atlas).

PROJECTS & CYBERSECURITY

- **Abyss Platform (ReactJS, Node.js, Python):** Built the web platform hosting AI tournament simulations for 200+ engineers to evaluate intelligent agents (MCTS, Minimax).
- **Perkastor (TypeScript, ReactJS, Python):** Developed an automated NLP pipeline scraping over 500,000 Wikipedia data points using Semantic Web techniques for historical mapping.
- **PolyCyber & CTF Competitions:** Vice-President of Communications for Polytechnique's Cybersecurity Committee. Ranked **4th/400** at United CTF Montreal and **4th/40** at Hackfest CTF Canada.